

Unit 10: Singular values and least squares

Singular values and orthogonal coordinates

All matrices in this unit are real. A matrix A of size $m \times n$ may be rectangular or singular. The symmetric spectral theorem applies to its Gram matrices $A^\top A$ and $AA^\top$, since $x^\top A^\top Ax = \|Ax\|^2 \geq 0$. Moreover,

$$\ker(A^\top A) = \ker A : \quad A^\top Ax = 0 \implies \|Ax\|^2 = 0 \implies Ax = 0,$$

and the reverse inclusion follows by multiplication. Thus $\text{rank}(A^\top A) = \text{rank } A$. Apply the same argument to $A^\top$ for $AA^\top$.

Theorem 10.1 (Singular value decomposition). Let A have rank $r > 0$. There are orthonormal lists $u_1, \dots, u_r \in \mathbb{R}^m$, $v_1, \dots, v_r \in \mathbb{R}^n$, and numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$ such that

$$Av_i = \sigma_i u_i, \quad A^\top u_i = \sigma_i v_i, \quad A = \sum_{i=1}^r \sigma_i u_i v_i^\top.$$

Writing $U_r = (u_1, \dots, u_r)$, $V_r = (v_1, \dots, v_r)$, and $D_r = \text{diag}(\sigma_1, \dots, \sigma_r)$, this is $A = U_r D_r V_r^\top$, with $U_r^\top U_r = V_r^\top V_r = I_r$. Extending both lists to orthonormal bases gives $A = U \Sigma V^\top$, where U, V are square orthogonal matrices and the only nonzero entries of the $m \times n$ matrix Σ are its first r diagonal entries σ_i . For $A = 0$, use $\Sigma = 0$ and any orthogonal U, V .

Proof. Choose an orthonormal eigenbasis $v_1, \dots, v_n$ of $A^\top A$, putting its r positive eigenvalues λ_i first, in decreasing order. Set $\sigma_i = \sqrt{\lambda_i}$ and $u_i = Av_i/\sigma_i$. Then

$$u_i^\top u_j = \frac{v_i^\top A^\top A v_j}{\sigma_i \sigma_j} = \frac{\lambda_j v_i^\top v_j}{\sigma_i \sigma_j} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Also $A^\top u_i = \lambda_i v_i/\sigma_i = \sigma_i v_i$. For $i > r$, $v_i \in \ker(A^\top A) = \ker A$, so $Av_i = 0$. Expanding an arbitrary x in the v_i basis gives

$$Ax = \sum_{i=1}^r \sigma_i u_i (v_i^\top x),$$

proving the matrix identity. Extend the u_i to an orthonormal basis of $\mathbb{R}^m$. The identities for Av_i say exactly that $U^\top AV = \Sigma$, hence $A = U \Sigma V^\top$. $\square$

The σ_i are the positive **singular values** of A . When all singular values are listed, append zeros to make $\min(m, n)$ entries. The eigenvalues of $A^\top A$ include these squares and, when necessary, further zeros to make n entries. The positive eigenvalues of $A^\top A$ and $AA^\top$ agree, including multiplicities: the constructed u_i are orthonormal eigenvectors of $AA^\top$, and its rank is also r .

The factors on the two sides are linked. Choose eigenvectors of $A^\top A$, then define $u_i = Av_i/\sigma_i$. Independently choosing signs or bases in both Gram eigenspaces can break that identity. For example, $A = (1)$, $u_1 = 1$, $v_1 = -1$ satisfy both Gram eigenvalue equations but give $u_1 \sigma_1 v_1^\top = -1$, not A . When a positive singular value repeats, the same orthogonal change must be made in its left and right singular-vector lists.

Example 10.1 (A rectangular decomposition). Let

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^\top A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

In the plane of first and third coordinates, $(1, 0, 1)^\top$ and $(-1, 0, 1)^\top$ have eigenvalues 2, 0; e_2 has eigenvalue 1. Normalize in the order 2, 1, 0:

$$v_1 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top, \quad v_2 = (0, 1, 0)^\top, \quad v_3 = \frac{1}{\sqrt{2}}(-1, 0, 1)^\top.$$

The positive singular values are $\sqrt{2}$, 1, and

$$u_1 = Av_1/\sqrt{2} = (1, 0)^\top, \quad u_2 = Av_2 = (0, 1)^\top.$$

Thus $U = I_2$,

$$V = \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad U\Sigma V^\top = A.$$

The first row of $\Sigma V^\top$ is $(1, 0, 1)$, and the second is $(0, 1, 0)$, verifying reconstruction. The formula $Ax = \sqrt{2}u_1(v_1^\top x) + u_2(v_2^\top x)$ shows which direction is lost and which directions are stretched.

The four subspaces and the size of a matrix

The SVD gives orthonormal bases for the four fundamental subspaces:

$$\begin{aligned} \operatorname{im} A &= \operatorname{span}(u_1, \dots, u_r), & \ker A^\top &= \operatorname{span}(u_{r+1}, \dots, u_m), \\ \operatorname{im} A^\top &= \operatorname{span}(v_1, \dots, v_r), & \ker A &= \operatorname{span}(v_{r+1}, \dots, v_n). \end{aligned}$$

The expansion of Ax gives the image inclusion, and $u_i = A(v_i/\sigma_i)$ gives its reverse. Independence of the u_i makes $Ax = 0$ equivalent to $v_i^\top x = 0$ for $i \leq r$. Apply the same argument to $A^\top$. In particular A restricts to a bijection $\operatorname{im} A^\top \rightarrow \operatorname{im} A$.

For $x = \sum_i z_i v_i$, orthogonality gives $\|Ax\|^2 = \sum_{i=1}^r \sigma_i^2 z_i^2$. Define $\|A\|_2 = \max_{\|x\|=1} \|Ax\|$, the operator norm. For $r > 0$, this identity gives $\|A\|_2 = \sigma_1$, attained at v_1 . On $\operatorname{im} A^\top$, it also gives

$$\sigma_r \|x\| \leq \|Ax\| \leq \sigma_1 \|x\|.$$

The Frobenius norm satisfies

$$\|A\|_F^2 = \operatorname{tr}(A^\top A) = \sum_{i=1}^r \sigma_i^2.$$

Indeed $A^\top A = V_r D_r^2 V_r^\top$ and cyclicity of trace give $\operatorname{tr}(A^\top A) = \operatorname{tr}(D_r^2 V_r^\top V_r) = \operatorname{tr}(D_r^2)$.

The Moore–Penrose inverse

The preceding bijection has an inverse on the image. Extend it by zero on the perpendicular complement of the image. Its matrix is

$$A^+ = V_r D_r^{-1} U_r^\top = \sum_{i=1}^r \frac{1}{\sigma_i} v_i u_i^\top, \quad 0^+ = 0_{n \times m}.$$

This **Moore–Penrose inverse** has size $n \times m$. For nonsingular square A , it is the ordinary inverse. For all A , multiplication gives

$$AA^+ = U_r U_r^\top =: P, \quad A^+A = V_r V_r^\top =: Q.$$

Thus P projects orthogonally onto $\operatorname{im} A$, and Q onto $\operatorname{im} A^\top$. They generally differ, possibly in size.

Theorem 10.2 (Four equations and uniqueness). The matrix A^+ is the unique $n \times m$ matrix X satisfying

$$AXA = A, \quad XAX = X, \quad (AX)^\top = AX, \quad (XA)^\top = XA.$$

Consequently its definition is independent of the chosen SVD.

Proof. The SVD formula gives symmetric $AA^+ = P$, $A^+A = Q$, and

$$AA^+A = U_r U_r^\top U_r D_r V_r^\top = A, \quad A^+AA^+ = V_r V_r^\top V_r D_r^{-1} U_r^\top = A^+.$$

For uniqueness, suppose X satisfies the four equations. The matrix AX is symmetric idempotent, and its image is $\text{im } A$: one inclusion follows from its factor A , and $AX(Az) = Az$ gives the reverse. Hence $AX = P$. Likewise XA is symmetric idempotent and $\ker(XA) = \ker A$, since $XAz = 0$ implies $Az = AXAz = 0$. Its image is therefore $(\ker A)^\perp = \text{im } A^\top$, so $XA = Q$. Finally

$$X = XAX = QX = A^+AX = A^+P = A^+.$$

Here uniqueness of the orthogonal projector follows from uniqueness of the orthogonal decomposition. $\square$

Example 10.2 (Recovering the part that can be recovered). For the rectangular matrix of Example 10.1,

$$A^+ = \frac{1}{\sqrt{2}}v_1u_1^\top + v_2u_2^\top = \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \\ 1/2 & 0 \end{pmatrix}, \quad AA^+ = I_2, \quad A^+A = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

For $b = (2, 3)^\top$, $x_0 = A^+b = (1, 3, 1)^\top$ satisfies $Ax_0 = b$. All exact solutions are

$$x = (1, 3, 1)^\top + t(-1, 0, 1)^\top, \quad t \in \mathbb{R},$$

since $(-1, 0, 1)^\top$ spans the kernel. The two terms are orthogonal, so $\|x\|^2 = 11 + 2t^2$. The inverse selects the unique solution of smallest norm, though there are infinitely many exact solutions.

Least squares as orthogonal projection

Given $b \in \mathbb{R}^m$, least squares asks for a vector x minimizing $\|Ax - b\|^2$. Set

$$p = Pb = AA^+b, \quad r = b - p, \quad x_0 = A^+b.$$

Here $p \in \text{im } A$, $r \perp \text{im } A$, and $Ax_0 = p$. For every x , Pythagoras therefore gives

$$Ax - b = A(x - x_0) - r, \quad \|Ax - b\|^2 = \|A(x - x_0)\|^2 + \|r\|^2.$$

This proves existence without assuming full rank or consistency.

Theorem 10.3 (All least-squares solutions). For any real $m \times n$ matrix A and any $b \in \mathbb{R}^m$, the following are equivalent:

- (a) x minimizes $\|Ax - b\|^2$;
- (b) $Ax = AA^+b$;
- (c) $A^\top Ax = A^\top b$, the **normal equations**;
- (d) $x = A^+b + z$ for some $z \in \ker A$, equivalently $x = A^+b + (I_n - A^+A)w$ for arbitrary $w \in \mathbb{R}^n$.

All minimizers give the same fitted vector p and residual r . The unique minimizer of smallest norm is $x_0 = A^+b$. The exact system $Ax = b$ is consistent if and only if $Pb = b$.

Proof. The squared-distance identity proves (a) $\Leftrightarrow$ (b) and (b) $\Leftrightarrow A(x - x_0) = 0$, which gives the first form in (d). The projector $I - Q$ has image $\ker A$, giving the second form. Condition (b) says $b - Ax \perp \operatorname{im} A$, equivalent to $A^\top(b - Ax) = 0$, which is (c). Conversely (c) gives that same orthogonality, hence the unique projection, proving (b). Since $x_0 \in \operatorname{im} A^\top = (\ker A)^\perp$,

$$\|x_0 + z\|^2 = \|x_0\|^2 + \|z\|^2 \quad (z \in \ker A),$$

with equality to $\|x_0\|^2$ only for $z = 0$. Finally $Ax = b$ is solvable exactly when $b \in \operatorname{im} A$, exactly when its projection is itself. $\square$

If A has full column rank, $A^\top A$ is positive definite because $\|Ax\|^2 > 0$ for nonzero x . The normal equations then give

$$x = (A^\top A)^{-1} A^\top b, \quad A^+ = (A^\top A)^{-1} A^\top.$$

Applying this to $A^\top$ when A has full row rank gives $A^+ = A^\top (AA^\top)^{-1}$. Then $AA^+ = I_m$, so every right-hand side has an exact solution. Without full column rank, uniqueness of the fitted vector differs from uniqueness of its coefficients.

Example 10.3 (An inconsistent system with dependent columns). Let

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Both rows of Ax are equal, so the system is inconsistent. Compute

$$A^\top A = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}, \quad v_1 = \frac{1}{\sqrt{5}}(1, 2)^\top, \quad v_2 = \frac{1}{\sqrt{5}}(-2, 1)^\top.$$

Their eigenvalues are 10, 0. Thus $\sigma_1 = \sqrt{10}$, $u_1 = Av_1/\sqrt{10} = (1, 1)^\top/\sqrt{2}$; take $u_2 = (-1, 1)^\top/\sqrt{2}$. It follows that

$$A = \sqrt{10} u_1 v_1^\top, \quad A^+ = \frac{1}{10} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \quad P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad Q = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Multiplying gives

$$x_0 = \begin{pmatrix} 3/10 \\ 3/5 \end{pmatrix}, \quad p = \begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}, \quad r = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}.$$

The normal equations are $2x_1 + 4x_2 = 3$, $4x_1 + 8x_2 = 6$, so all least-squares solutions are

$$x = \begin{pmatrix} 3/10 \\ 3/5 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Each fits p and has residual sum of squares $1/2$. Its squared norm is $9/20 + 5t^2$, showing the minimum-norm choice. For arbitrary coefficients the entire objective is

$$\|Ax - b\|^2 = (x_1 + 2x_2 - 1)^2 + (x_1 + 2x_2 - 2)^2 = 2(x_1 + 2x_2 - 3/2)^2 + 1/2.$$

Thus projection, SVD, and the normal equations agree.

Sensitivity and small singular values

For fixed nonzero A , changing the right-hand side by h changes its minimum-norm least-squares solution by

$$A^+h = \sum_{i=1}^r \frac{u_i^\top h}{\sigma_i} v_i.$$

Bessel's inequality gives

$$\|A^+h\|^2 = \sum_{i=1}^r \frac{(u_i^\top h)^2}{\sigma_i^2} \leq \frac{1}{\sigma_r^2} \|h\|^2.$$

Equality holds for $h = u_r$, so $\|A^+\|_2 = 1/\sigma_r$. A small positive singular value can strongly amplify one component of an error in b . A component in $\ker A^\top$ changes the residual and has no effect on A^+b . For full column rank, the eigenvalues of $A^\top A$ range from σ_n^2 to σ_1^2 ; their ratio is the square of σ_1/σ_n . This explains why small singular values matter when calculating a solution, even though the normal equations give an exact theoretical characterization.

Exercises and solutions

All matrices are real and all displayed products are conformable. Matrix dimensions are positive integers. Compact SVD factors have orthonormal columns. The final two subsections extend the vector results to matrix equations and generalized inverses.

Suggested practice: 10.2, 10.6, 10.7, 10.9, 10.10, 10.12, 10.21, 10.24, 10.25, 10.29, 10.35, 10.37, 10.41, 10.42.

Constructing and interpreting singular values

Exercise 10.1. For a real $m \times n$ matrix A of rank $r > 0$, construct matrices $S \in \mathbb{R}^{m \times r}$, $T \in \mathbb{R}^{n \times r}$, with orthonormal columns, and a positive diagonal $r \times r$ matrix Λ such that $A = S\Lambda^{1/2}T^\top$. Start with the eigenvectors of $AA^\top$.

Solution. Choose orthonormal eigenvectors $s_1, \dots, s_r$ for the positive eigenvalues of $AA^\top$, and let S have these columns and Λ these eigenvalues. Define $T = A^\top S\Lambda^{-1/2}$. Then

$$T^\top T = \Lambda^{-1/2} S^\top A A^\top S \Lambda^{-1/2} = I_r.$$

The remaining orthonormal eigenvectors form S_0 ; for each such vector z , $\|A^\top z\|^2 = z^\top A A^\top z = 0$, so $S_0^\top A = 0$. Consequently

$$A = (SS^\top + S_0 S_0^\top)A = SS^\top A = S\Lambda^{1/2}T^\top.$$

Exercise 10.2. In an SVD, the equations $AA^\top S = S\Lambda$ and $A^\top AT = T\Lambda$ hold. Explain why solving these two equations independently, even with orthonormal columns, need not give $A = S\Lambda^{1/2}T^\top$. Give the correct procedure.

Solution. The equations do not link the two choices of signs or repeated-eigenspace bases. For $A = (1)$, $S = (1)$, $T = (-1)$, $\Lambda = (1)$, both eigenvalue equations hold but the proposed product is -1 . After choosing S, Λ from $AA^\top$, set $T = A^\top S\Lambda^{-1/2}$, as in the preceding proof. Alternatively choose T first and set $S = AT\Lambda^{-1/2}$. For a repeated positive singular value the same orthogonal change must be applied to both corresponding lists.

Exercise 10.3. Prove the two full-rank polar decompositions: (a) if $A \in \mathbb{R}^{m \times n}$ has full column rank, then $A = PH$ with $P^\top P = I_n$ and H positive definite; (b) if A has full row rank, then $A = HP^\top$ with $P^\top P = I_m$ and H positive definite.

Solution. (a) Set $H = (A^\top A)^{1/2}$, positive definite by full column rank, and $P = AH^{-1}$. Then $PH = A$ and $P^\top P = H^{-1}A^\top AH^{-1} = I_n$. (b) Apply (a) to $A^\top$: write $A^\top = PH$, where now $H = (AA^\top)^{1/2}$ and $P^\top P = I_m$. Transposition gives $A = HP^\top$, since $H = H^\top$.

Exercise 10.4. Let A, B be symmetric, with A, B , and $A - B$ positive semidefinite. If A is singular, prove that B and $A - B$ are singular.

Solution. Choose nonzero $z \in \ker A$. Then $0 = z^\top Az = z^\top Bz + z^\top (A - B)z$, a sum of nonnegative terms, so both vanish. For a positive semidefinite C , $z^\top Cz = \|C^{1/2}z\|^2 = 0$ implies $Cz = 0$. Apply this to $B, A - B$; both have the nonzero kernel vector z .

Exercise 10.5. For a positive semidefinite A , prove that $A^{1/2}x, A^{1/2}y$ are dependent if and only if Ax, Ay are dependent.

Solution. An orthonormal eigenbasis shows $\ker A = \ker A^{1/2}$, since a nonnegative eigenvalue and its square root vanish together. For any $(a, b) \neq (0, 0)$,

$$aAx + bAy = 0 \iff ax + by \in \ker A \iff ax + by \in \ker A^{1/2} \iff aA^{1/2}x + bA^{1/2}y = 0.$$

Thus a nontrivial relation exists for one pair exactly when it exists for the other.

Exercise 10.6. Find all eigenvalues, including nonreal ones, and singular values of

$$(a) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (b) \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad (c) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (d) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Solution. (a) The characteristic polynomial is t^2 , so both eigenvalues are zero; $A^\top A = \text{diag}(0, 1)$, giving singular values 1, 0. (b) The triangular matrix has eigenvalues 1, 0; $A^\top A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has eigenvectors $(1, 1)^\top, (1, -1)^\top$ with eigenvalues 2, 0. The singular values are $\sqrt{2}, 0$. (c) Those same directions give eigenvalues 2, 0 for A , and $A^\top A = 2A$, so its eigenvalues are 4, 0. The singular values are 2, 0. (d) The characteristic polynomial factors as $t^2 + 1 = (t - i)(t + i)$, so the eigenvalues are $i, -i$. For example, $A(1, i)^\top = (i, -1)^\top = i(1, i)^\top$; conjugating gives the other eigenvector. Since $A^\top A = I_2$, both singular values are 1. Thus the nonreal eigenvalues of this real rotation coexist with real, nonnegative singular values.

Exercise 10.7. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix}.$$

Find the eigenvalues and singular values of A and $A^\top$. For $A^\top$, list the eigenvalues, their absolute values, and singular values in decreasing order.

Solution. Block triangularity gives the characteristic polynomial $t(t - 2)(t + 1)$. Transposition preserves this polynomial. Furthermore,

$$A^\top A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 1 \end{pmatrix}.$$

The mutually orthogonal vectors $(1, 1, 0)^\top, (1, -1, -1)^\top, (1, -1, 2)^\top$ give eigenvalues 4, 3, 0, respectively, by direct multiplication. Thus both A and $A^\top$ have singular values $2, \sqrt{3}, 0$. The three decreasing lists are 2, 0, -1; 2, 1, 0; and 2, $\sqrt{3}$, 0.

Exercise 10.8. For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, compute the eigenvalues, singular values, a spectral decomposition, and the modulus $|A| = (A^\top A)^{1/2}$.

Solution. The orthonormal vectors $q_1 = (1, 1)^\top/\sqrt{2}, q_2 = (1, -1)^\top/\sqrt{2}$ have eigenvalues 3, -1, so $A = 3q_1q_1^\top - q_2q_2^\top$. Squaring gives $A^\top A = 9q_1q_1^\top + q_2q_2^\top$, with singular values 3, 1. Its positive semidefinite square root is

$$|A| = 3q_1q_1^\top + q_2q_2^\top = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The negative eigenvalue of A becomes a positive singular value.

Exercise 10.9. Find a Schur triangular form (an upper triangular matrix obtained by orthogonal similarity) and a full SVD for

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Solution. The orthonormal basis $(e_1 + e_4)/\sqrt{2}, (e_1 - e_4)/\sqrt{2}, e_2, e_3$ gives the upper triangular matrix $\text{diag}\left(2, 0, \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}\right)$. Indeed the first two basis vectors have eigenvalues 2, 0, while $Ae_2 = 2e_2$ and $Ae_3 = 3e_2 + 2e_3$. For the remaining 2×2 block, its Gram matrix is $\begin{pmatrix} 4 & 6 \\ 6 & 13 \end{pmatrix}$; multiplying $(1, 2)^\top$ and $(-2, 1)^\top$ gives eigenvalues 16, 1. Thus, in decreasing singular-value order, choose

$$\begin{aligned} v_1 &= (0, 1, 2, 0)^\top/\sqrt{5}, & u_1 &= (0, 2, 1, 0)^\top/\sqrt{5}, \\ v_2 &= (1, 0, 0, 1)^\top/\sqrt{2}, & u_2 &= v_2, \\ v_3 &= (0, -2, 1, 0)^\top/\sqrt{5}, & u_3 &= (0, -1, 2, 0)^\top/\sqrt{5}, \\ v_4 &= (1, 0, 0, -1)^\top/\sqrt{2}, & u_4 &= v_4. \end{aligned}$$

Both lists are orthonormal: their two coordinate planes are perpendicular, and the pairs within each plane have zero dot product and unit norm. Direct multiplication gives $Av_1 = 4u_1$, $Av_2 = 2u_2$, $Av_3 = u_3$, $Av_4 = 0$. Hence, with the displayed columns, $A = U \text{diag}(4, 2, 1, 0) V^\top$, the required full SVD.

Moore–Penrose inverses and their identities

Exercise 10.10. Find the Moore–Penrose inverse of (a) a nonsingular square matrix; (b) a scalar; (c) a diagonal matrix; (d) an $m \times n$ zero matrix.

Solution. (a) A^{-1} : both products are I , and the two triple products are A and A^{-1} . (b) For scalar a , use $a^+ = 1/a$ when $a \neq 0$, and zero when $a = 0$. Each scalar then satisfies the four equations. (c) For $D = \text{diag}(d_i)$, take $D^+ = \text{diag}(d_i^+)$. The triple equations hold coordinatewise; $DD^+ = D^+D$ is diagonal with entries zero or one, hence symmetric. (d) The inverse is $0_{n \times m}$. All four equations are zero equalities of the appropriate sizes.

Exercise 10.11. Prove existence of the Moore–Penrose inverse by checking all four equations.

Solution. If $A = U_r D V_r^\top$, put $X = V_r D^{-1} U_r^\top$. Then $AX = U_r U_r^\top$ and $XA = V_r V_r^\top$, both symmetric. Moreover $AXA = U_r U_r^\top U_r D V_r^\top = A$ and $XAX = V_r V_r^\top V_r D^{-1} U_r^\top = X$. When $A = 0$, the zero matrix of reversed size satisfies every equation.

Exercise 10.12. Prove uniqueness using only the four Moore–Penrose equations.

Solution. Let B, C both satisfy the equations for A . Symmetry and $ACA = A$ give

$$AB = B^\top A^\top = B^\top (ACA)^\top = (AB)^\top (AC)^\top = ABAC = AC,$$

where $ABA = A$ gives the last equality. Likewise

$$BA = A^\top B^\top = (ACA)^\top B^\top = (CA)^\top (BA)^\top = CABA = CA.$$

Now $B = BAB = (BA)B = (CA)B = C(AB) = C(AC) = CAC = C$.

Exercise 10.13. Prove $(A^+)^+ = A$ and $(A^\top)^+ = (A^+)^\top$.

Solution. The four equations are unchanged when A, A^+ are interchanged: the triple equations exchange roles, and the two symmetry requirements exchange roles. Uniqueness therefore gives $(A^+)^+ = A$. Transposing $AA^+A = A$ and $A^+AA^+ = A^+$ gives the two triple equations for $A^\top, (A^+)^\top$. Their products are $(A^+A)^\top$ and $(AA^+)^\top$, which are symmetric. Uniqueness proves the second identity.

Exercise 10.14. For $A \in \mathbb{R}^{m \times n}$, show that all six matrices are symmetric idempotent:

$$AA^+, \quad A^+A, \quad I_m - AA^+, \quad I_n - A^+A, \quad A(A^\top A)^+A^\top, \quad A^\top(AA^\top)^+A.$$

Solution. Write $A = UDV^\top$ with orthonormal columns in the compact factors. Then the first two matrices are $UU^\top, VV^\top$, whose squares are themselves. Their complements square to themselves because $(I - P)^2 = I - 2P + P^2 = I - P$, and are symmetric. Also $(A^\top A)^+ = VD^{-2}V^\top$, since the four equations reduce to the nonzero diagonal entries. Thus $A(A^\top A)^+A^\top = UU^\top$. Similarly the sixth matrix is $VV^\top$. The zero case gives zero projectors and identity complements.

Exercise 10.15. (a) If A is symmetric idempotent, prove $A^+ = A$. (b) Does the converse hold? (c) Show $(AA^+)^+ = AA^+$ and $(A^+A)^+ = A^+A$.

Solution. (a) The choice $X = A$ gives both triple products $A^3 = A$, and both products are the symmetric matrix $A^2 = A$; uniqueness proves the claim. (b) No: for $A = -I$, the inverse is $-I = A$, but $A^2 = I \neq A$. (c) Both matrices are symmetric idempotent by the preceding exercise, so apply (a) to each.

Exercise 10.16. Show that A, A^+, AA^+, A^+A have the same rank.

Solution. For compact $A = UDV^\top$ of rank r , the matrix $A^+ = VD^{-1}U^\top$ has image $\text{im } V$, of dimension r . The matrices $AA^+ = UU^\top$ and $A^+A = VV^\top$ project onto r -dimensional spaces, so also have rank r . For $A = 0$ every rank is zero.

Exercise 10.17. Writing $A^{+\top} = (A^+)^\top$, prove

- (a) $A^\top AA^+ = A^\top = A^+ AA^\top$;
- (b) $A^\top A^{+\top} A^+ = A^+ = A^+ A^{+\top} A^\top$;
- (c) $(A^\top A)^+ = A^+ A^{+\top}$;
- (d) $(AA^\top)^+ = A^{+\top} A^+$;
- (e) $A^+ = (A^\top A)^+ A^\top = A^\top (AA^\top)^+$.

Solution. Let $A = UDV^\top$, $A^+ = VD^{-1}U^\top$, with $U^\top U = V^\top V = I$. (a) Both products reduce to $VDU^\top = A^\top$. (b) Both products reduce to $VD^{-1}U^\top = A^+$. (c) $A^\top A = VD^2V^\top$, whose MP-inverse is $VD^{-2}V^\top$; also $A^+ A^{+\top} = VD^{-1}U^\top U D^{-1}V^\top = VD^{-2}V^\top$. (d) The same multiplication gives $(AA^\top)^+ = UD^{-2}U^\top = A^{+\top} A^+$. (e) Multiplying the expressions in (c),(d) by $A^\top = VDU^\top$ on the stated sides gives $VD^{-1}U^\top$ in both cases. For $A = 0$, all displayed products vanish.

Exercise 10.18. Prove the following identities, with all products conformable:

- (a) $A(A^\top A)^+ A^\top A = A = AA^\top (AA^\top)^+ A$.
- (b) For positive definite V , $(X^\top V^{-1} X)(X^\top V^{-1} X)^+ X^\top = X^\top$.
- (c) Set $C = (B^\top B)^+$ and $K = ABCB^\top A^\top$. Show $KK^+ AB = AB$.

Solution. (a) Each left expression is $AA^+A = A$, by the identities just proved. (b) Put $F = V^{-1/2}X$, $G = F^\top F = X^\top V^{-1}X$. Part (a) gives $FG^+G = F$. Multiply by $V^{1/2}$ and transpose: $GG^+X^\top = X^\top$, since G, G^+ are symmetric. (c) The matrix C is positive semidefinite; let $F = C^{1/2}$ and $D = ABF$. Then $DD^\top = K$, and (a) gives $KK^+D = D$. Multiply on the right by $FB^\top B$. Since $BCB^\top B = B$, also by (a), we have $DFB^\top B = ABCB^\top B = AB$. Thus $KK^+AB = AB$, as required.

Exercise 10.19. Prove $A^+ = (A^\top A)^{-1}A^\top$ for full column rank and $A^+ = A^\top(AA^\top)^{-1}$ for full row rank. Deduce $A^+A = I_n$ in the first case and $AA^+ = I_m$ in the second.

Solution. Full column rank gives $x^\top A^\top A x = \|Ax\|^2 > 0$ for nonzero x , so $A^\top A$ is nonsingular. Its MP-inverse is its ordinary inverse, and the identity $A^+ = (A^\top A)^+ A^\top$ gives the formula. Apply the same argument to $A^\top$, using $(A^\top)^+ = (A^+)^^\top$, for the full row rank case. Multiplication by A on the indicated side gives the two identities.

Exercise 10.20. Prove (a) $A = 0 \iff A^+ = 0$; (b) $AB = 0 \iff B^+A^+ = 0$; (c) $A^+B = 0 \iff A^\top B = 0$.

Solution. (a) Zero has zero MP-inverse; the reverse follows from $A = AA^+A$. (b) If $AB = 0$, then

$$B^+A^+ = (B^\top B)^+ B^\top A^\top (AA^\top)^+ = 0.$$

Apply this implication to the pair B^+, A^+ and use $(A^+)^+ = A$, $(B^+)^+ = B$ for the reverse. (c) If $A^\top B = 0$, then $A^+B = (A^\top A)^+ A^\top B = 0$. Conversely $A^\top B = A^\top AA^+B = 0$.

Exercise 10.21. If $\text{rank } A = 1$, prove $A^+ = A^\top / \text{tr}(AA^\top)$. In particular determine the MP-inverse of a column vector, including the zero vector.

Solution. Write $A = ab^\top$ with both vectors nonzero. Set $c = \|a\|^2 \|b\|^2 = \text{tr}(AA^\top) > 0$, $X = ba^\top / c$. Then $AX = aa^\top / \|a\|^2$ and $XA = bb^\top / \|b\|^2$, both symmetric. They act as the identity on a , respectively b , so $AXA = A$ and $XAX = X$. Hence $X = A^+$. Taking scalar $b = 1$ gives $a^+ = a^\top / (a^\top a)$ for $a \neq 0$. For a zero m -column the inverse is a zero $1 \times m$ row.

Exercise 10.22. Prove that the MP-inverse of $\text{diag}(A_1, A_2)$ is $\text{diag}(A_1^+, A_2^+)$, allowing rectangular diagonal blocks.

Solution. Call the two displayed block matrices A, X . Block multiplication gives $AXA = \text{diag}(A_1A_1^+A_1, A_2A_2^+A_2) = A$ and $XAX = X$. The products AX and XA are block diagonal with symmetric blocks $A_iA_i^+$ and $A_i^+A_i$, respectively. Thus all four equations hold, and uniqueness identifies $X = A^+$.

Exercise 10.23. Let $A = SAS^\top$ be symmetric of rank $r > 0$, where $S^\top S = I_r$ and Λ is diagonal with nonzero entries, possibly negative. Prove $A^+ = S\Lambda^{-1}S^\top$.

Solution. Put $X = S\Lambda^{-1}S^\top$. Both products AX, XA equal $SS^\top$, a symmetric matrix. Also $AXA = SS^\top SAS^\top = A$ and $XAX = SS^\top S\Lambda^{-1}S^\top = X$. Thus X satisfies all four equations, regardless of the signs of the nonzero diagonal entries.

Products, projections, and ranges

Exercise 10.24. Prove the following reverse-order formulas:

- (a) If $BB^\top = I$, then $(AB)^+ = B^\top A^+$.
- (b) If A has full column rank and B full row rank, then $(AB)^+ = B^+ A^+$.
- (c) If $A = BC^\top$, with B, C both having $r > 0$ independent columns, then

$$A^+ = C(B^\top BC^\top C)^{-1} B^\top.$$

Solution. (a) Put $X = B^\top A^+$. The triple products reduce to $ABXAB = AA^+AB = AB$ and $XABX = B^\top A^+AA^+ = X$. The two products are $ABX = AA^+$ and $XAB = B^\top A^+AB$, both symmetric; thus $X = (AB)^+$. (b) Put $X = B^+ A^+$. Since $A^+A = I$, $BB^+ = I$, the triple products give $ABXAB = AB$, $XABX = X$. The two products reduce to AA^+ and B^+B , both symmetric. (c) Apply (b) to $B, C^\top$ and the full-rank formulas:

$$A^+ = C(C^\top C)^{-1}(B^\top B)^{-1}B^\top = C(B^\top BC^\top C)^{-1}B^\top.$$

Both Gram matrices are positive definite, so their inverses exist.

Exercise 10.25. Let A be square. (a) If A is symmetric, show $AA^+ = A^+A$. (b) Does the converse hold? (c) Prove that equality holds exactly when $\text{im } A = \text{im } A^\top$.

Solution. (a) In a compact spectral decomposition $A = SAS^\top$, both products are $SS^\top$; the zero case is immediate. (b) No: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is nonsymmetric and nonsingular, so both products are I_2 . (c) The two products are the orthogonal projectors onto the two indicated images. Equality of matrices implies equality of their images. Conversely, if the images agree, uniqueness of the orthogonal projection onto that common subspace gives equality of the matrices.

Exercise 10.26. If $A = BB^\top$, where B has $r > 0$ independent columns, prove $A^+ = B(B^\top B)^{-2}B^\top$.

Solution. The full-rank reverse-order formula for $BB^\top$ gives

$$A^+ = (B^\top)^+ B^+ = B(B^\top B)^{-1}(B^\top B)^{-1}B^\top.$$

Here $B^\top B$ is positive definite, so the stated expression exists. For a rank-zero positive semidefinite matrix, the separate answer is $A^+ = 0$.

Exercise 10.27. Prove $A^\top AB = A^\top C \iff AB = AA^+C$.

Solution. If the first equality holds, premultiply by $A(A^\top A)^+$:

$$AB = A(A^\top A)^+ A^\top AB = A(A^\top A)^+ A^\top C = AA^+C.$$

The two simplifications use the preceding MP identities. Conversely multiply $AB = AA^+C$ by $A^\top$ and use $A^\top AA^+ = A^\top$. This gives $A^\top AB = A^\top C$.

Exercise 10.28. If B has full row rank, prove $(AB)(AB)^+ = AA^+$. Is full row rank necessary for this equality?

Solution. The inclusion $\text{im}(AB) \subseteq \text{im } A$ always holds. Full row rank gives $BB^+ = I$, hence $A = (AB)B^+$, which gives the reverse inclusion. The two orthogonal projectors therefore agree. The hypothesis is not necessary: let $A = B = \text{diag}(1, 0)$. Then $AB = A$ and both projectors equal A , although B has rank one rather than two.

Exercise 10.29. Let A be symmetric idempotent. (a) If $AB = B$, show that $A - BB^+$ is symmetric idempotent of rank $\text{rank } A - \text{rank } B$. (b) Does the converse hold?

Solution. Put $P = BB^+$. From $AB = B$, $AP = P$, and transposition gives $PA = P$. Therefore $(A - P)^2 = A - AP - PA + P = A - P$, and symmetry is immediate. Its image is $\text{im } A \cap (\text{im } B)^\perp$: $A(A - P) = A - P$, $P(A - P) = 0$; conversely $Az = z$, $Pz = 0$ gives $(A - P)z = z$. Since $\text{im } B \subseteq \text{im } A$, orthogonal decomposition within $\text{im } A$ gives the rank difference. (b) Yes; idempotence already suffices. If $C = A - P$ is idempotent, it is symmetric, so $z^\top Cz = \|Cz\|^2 \geq 0$. For $z \in \text{im } P$, this says $z^\top Az \geq \|z\|^2$. But A is an orthogonal projector, so $z^\top Az = \|Az\|^2 \leq \|z\|^2$. Equality and Pythagoras imply $(I - A)z = 0$. Thus A fixes every column of B , giving $AB = B$.

Exercise 10.30. For a symmetric idempotent $n \times n$ matrix A , show: (a) if $AB = B$ and $\text{rank } A = \text{rank } B$, then $A = BB^+$; (b) if $AB = 0$ and $\text{rank } A + \text{rank } B = n$, then $A = I_n - BB^+$.

Solution. (a) The preceding exercise makes $A - BB^+$ a matrix of rank zero, so it vanishes. (b) The matrix $C = I - A$ is symmetric idempotent and fixes B . Its image is $\ker A$, so its rank is $n - \text{rank } A = \text{rank } B$. Part (a) applied to C gives $I - A = BB^+$.

Vector equations and least squares

Exercise 10.31. (a) Show that every solution of $Ax = 0$ has the form $x = (I - A^+A)q$, with q arbitrary, and that every such vector is a solution. (b) Determine when the solution is unique. (c) Prove that otherwise there are infinitely many solutions.

Solution. Multiplication gives $A(I - A^+A)q = (A - AA^+A)q = 0$. Conversely if $Ax = 0$, then $(I - A^+A)x = x$, so choose $q = x$. The solution is unique exactly when $\ker A = \{0\}$, exactly when A has full column rank; it is then zero. Otherwise choose nonzero $z \in \ker A$. The vectors tz , $t \in \mathbb{R}$, are distinct solutions, proving infinitude.

Exercise 10.32. Give a nonhomogeneous real system $Ax = b$, $b \neq 0$, with no solution.

Solution. Take $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$, $b = (1, 2)^\top$. The two equations would require the same quantity $x_1 + 2x_2$ to equal both one and two, a contradiction.

Exercise 10.33. (a) Show that the following are equivalent: $Ax = b$ is consistent; $b \in \text{im } A$; $\text{rank}(A \mid b) = \text{rank } A$; and $AA^+b = b$.

(b) Does full column rank of A guarantee consistency of $Ax = b$? Give an explicit counterexample if it does not.

Solution. (a) The first two are equivalent because Ax runs through the column space of A . Appending b leaves the column-space dimension unchanged exactly when b was already in that space; otherwise it increases the dimension by one. This proves the third equivalence. Finally AA^+ projects onto that space, and fixes exactly its members. Equivalently, consistency gives $b = Ax = AA^+Ax = AA^+b$, while $AA^+b = b$ supplies the solution $x = A^+b$.

(b) No. Take $A = (1, 0)^\top$ and $b = (0, 1)^\top$. The single column of A is nonzero, so A has full column rank. But $Ax = (x, 0)^\top$ cannot equal b , whose second coordinate is one.

Exercise 10.34. If $Ax = b$ is consistent, prove that its complete solution set is $x = A^+b + (I - A^+A)q$, where q is arbitrary.

Solution. Consistency gives $AA^+b = b$, so $x_0 = A^+b$ is one solution. Then $Ax = b \iff A(x - x_0) = 0$. The complete kernel formula in Exercise 10.31 gives exactly $x - x_0 = (I - A^+A)q$. The same calculation verifies every proposed vector.

Exercise 10.35. Whether or not $Ax = b$ is consistent, prove

$$\|Ax - b\|^2 = (x - A^+b)^\top A^\top A(x - A^+b) + b^\top (I - AA^+)b.$$

Deduce that A^+b minimizes the residual norm.

Solution. Let $P = AA^+$. Write $Ax - b = A(x - A^+b) - (I - P)b$. The two terms are perpendicular, since $A^\top(I - P) = 0$. Their squared norms add. Because $I - P$ is symmetric idempotent, $\|(I - P)b\|^2 = b^\top(I - P)b$, proving the formula. The first term is $\|A(x - A^+b)\|^2 \geq 0$, and equals zero at $x = A^+b$. Thus this choice attains the global minimum.

Exercise 10.36. Prove (a) that $Ax = b$ is consistent for every $b \in \mathbb{R}^m$ exactly when A has full row rank; (b) that a consistent $Ax = b$ has a unique solution exactly when A has full column rank.

Solution. (a) Consistency for every b means $\text{im } A = \mathbb{R}^m$, equivalent to rank m , or full row rank. (b) Fix one solution x_0 . All solutions are $x_0 + \ker A$, so uniqueness is equivalent to $\ker A = \{0\}$. Rank-nullity makes this equivalent to rank n , or full column rank. Neither conclusion alone requires A to be square.

Matrix equations

Exercise 10.37. Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{p \times q}$, and $C \in \mathbb{R}^{m \times q}$. Prove that $AXB = C$ is consistent exactly when $AA^+CB^+B = C$. If consistent, prove that all solutions are

$$X = A^+CB^+ + Q - A^+AQBB^+, \quad Q \in \mathbb{R}^{n \times p}.$$

Solution. If $AXB = C$, multiplication gives $AA^+CB^+B = AA^+AXBB^+B = AXB = C$. Conversely that equality shows that $X_0 = A^+CB^+$ is a solution. For arbitrary Q , $A(Q - A^+AQBB^+)B = AQB - AA^+AQBB^+B = 0$, so every displayed matrix is a solution. If X is any solution, set $Y = X - X_0$, so $AYB = 0$. Then $A^+AYBB^+ = 0$, and choosing $Q = Y$ in the formula returns $X_0 + Y = X$. Thus no solutions are missing.

Exercise 10.38. For the same positive matrix dimensions, prove: (a) $AXB = C$ is consistent for every C exactly when A has full row rank and B full column rank; (b) when consistent, its solution is unique exactly when A has full column rank and B full row rank.

Solution. (a) The two full-rank conditions give $AA^+ = I_m$, $B^+B = I_q$, so the preceding consistency test holds for every C . Conversely, if A lacks full row rank, choose nonzero $u \in \ker A^\top$. Then $u^\top AXB = 0$, whereas $C = ue_1^\top$ has $u^\top C = \|u\|^2 e_1^\top \neq 0$. If B lacks full column rank, choose nonzero $v \in \ker B$. Every AXB annihilates v , but $C = e_1 v^\top$ does not. (b) If both stated ranks are full, multiplying a homogeneous equation $AYB = 0$ by A^+ and B^+ gives $Y = 0$, proving uniqueness. If A has a nonzero kernel vector z , then $Y = ze_1^\top \neq 0$ satisfies $AYB = 0$. If B lacks full row rank, choose nonzero $w \in \ker B^\top$; then $Y = e_1 w^\top \neq 0$ also satisfies it. Adding either such Y to a solution disproves uniqueness.

Exercise 10.39. Find all matrices solving (a) $AX = 0$; (b) $XA = 0$.

Solution. (a) All solutions are $X = (I - A^+A)Q$, with arbitrary conformable Q . Multiplication by A gives zero. Conversely $AX = 0$ implies $(I - A^+A)X = X$, so choose $Q = X$. (b) All solutions are $X = Q(I - AA^+)$. Each annihilates A , since $(I - AA^+)A = 0$. Conversely $XA = 0$ gives $X(I - AA^+) = X$.

Generalized inverses

Exercise 10.40. A generalized inverse of A is any matrix G satisfying $AGA = A$. Prove existence and show that all such matrices are

$$G = A^+ + Q - A^+AQAA^+, \quad Q \text{ arbitrary and conformable.}$$

Solution. The choice $G = A^+$ establishes existence. For the displayed formula,

$$AGA = AA^+A + AQA - AA^+AQAA^+A = A + AQA - AQA = A.$$

Conversely, if $AGA = A$, set $Q = G - A^+$. Then $AQA = 0$, so $A^+AQAA^+ = 0$, and the displayed expression becomes $A^+ + Q = G$. Thus it gives precisely all generalized inverses.

Exercise 10.41. For any generalized inverse G of A , prove that GA is idempotent. Show that it need not be symmetric. If it is symmetric, prove that $GA = A^+A$, so this product is unique.

Solution. The identity $AGA = A$ gives $(GA)^2 = G(AGA) = GA$. For a counterexample, take $A = \text{diag}(1, 0)$, $G = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Then $AGA = A$, but $GA = G$ is nonsymmetric. If GA is symmetric, it is an orthogonal projector. Also $\ker(GA) = \ker A$: one inclusion is immediate; if $GAx = 0$, then $Ax = AGAx = 0$. Thus it projects onto $(\ker A)^\perp$, exactly the image of the orthogonal projector A^+A . Uniqueness gives equality.

Exercise 10.42. Let G be any generalized inverse of $A^\top A$. Prove $AGA^\top = A(A^\top A)^+A^\top$. Conclude that this matrix is unique, symmetric, and idempotent.

Solution. For nonzero $A = UDV^\top$, the generalized-inverse equation is $VD^2V^\top GVD^2V^\top = VD^2V^\top$. Multiply by $V^\top$ on the left and V on the right to get $D^2(V^\top GV)D^2 = D^2$, hence $V^\top GV = D^{-2}$. Therefore

$$AGA^\top = UD(V^\top GV)DU^\top = UU^\top = A(A^\top A)^+A^\top.$$

This orthogonal projector is independent of G , symmetric, and idempotent. If $A = 0$, the expression is zero for every G , with the same conclusions.

Exercise 10.43. Let A be symmetric idempotent. (a) Show that all its generalized inverses are $A + Q - AQA$. (b) Show that $B = A + \alpha(I - A)$ is a generalized inverse for every α . (c) Prove that B is nonsingular for $\alpha \neq 0$. (d) Conclude that a generalized inverse need not have the same rank as A .

Solution. (a) In the general formula for generalized inverses, $A^+ = A$ and $A^2 = A$, giving $A + Q - AQA$. (b) Take $Q = \alpha I$; the formula gives the stated B . (c) Since $A(I - A) = (I - A)A = 0$, multiplication on either side gives

$$[A + \alpha(I - A)][A + \alpha^{-1}(I - A)] = A + (I - A) = I.$$

The factors commute, so the second is a two-sided inverse. (d) If A is singular and $\alpha \neq 0$, then B has full rank while A does not. For example $A = \text{diag}(1, 0)$, $\alpha = 1$ gives $B = I_2$.

Exercise 10.44. For any generalized inverse G of A , prove that $Ax = b$ is consistent exactly when $AGb = b$. When consistent, show that all solutions are $x = Gb + (I - GA)q$, q arbitrary.

Solution. If $b = Ax$, then $AGb = AGAx = Ax = b$. Conversely $AGb = b$ exhibits Gb as a solution. Under consistency,

$$A[Gb + (I - GA)q] = b + (A - AGA)q = b,$$

so every displayed vector is a solution. If x is any solution, choose $q = x$: $Gb + (I - GA)x = Gb + x - Gb = x$. Thus the parametrization is complete. No symmetry of AG or GA is required for these exact-system statements.